✅ 1. Competency #2: Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem

- Highest relevance.
- Students must first extract the total used space from the table (e.g., Music + Photos = 550 + 338 = 888 MB), and verify that together with Free space (112 MB), it sums to 1000 MB—consistent with a 1 GB capacity.
- When all content is copied to a new 2000 MB stick, the total data remains 888 MB, so the new stick's state must be:
  - Used: 888 MB (with Music and Photos in the same internal ratio)
  - Free: 2000 − 888 = 1112 MB
- Among the four chart options, only the one that correctly reflects "used ≈ 44%, free ≈ 56%" and maintains the original Music/Photos proportion is valid.
- If a student mistakenly assumes the new stick "fills up automatically" or that "proportions reset," they will choose incorrectly—precisely what this competency assesses: judging whether a visual/mathematical result makes sense in context.

✅ 2. Competency #1: Interpret a mathematical result back into the real-world context

- Highly relevant.
- The core action is "copying all content to a larger-capacity device." Students must understand:
  - Files do not change size when moved;
  - The new stick will have significantly more free space;
  - The absolute sizes remain unchanged, so their relative proportions decrease on the larger drive.
- For example, if Photos were 33.8% on the old stick, they become ~16.9% on the new one (338 / 2000). Students must translate this numerical relationship into visual sector sizes in a chart.
- This process exemplifies interpreting an abstract quantity (888 MB out of 2000 MB) as a tangible real-world state (the new USB drive appears "mostly empty").

✅ 3. Competency #5: Interpret mathematical outcomes in a variety of formats and compare representations

- Highly relevant.
- The problem provides tabular data (exact numbers) and multiple pie chart options (visual proportions).
- Students must:
  1. Read Music, Photos, and Free values (in MB) from the table;
  2. Compute their percentages relative to 2000 MB;
  3. Visually match and compare these percentages to the sector sizes in the four pie charts.
- This requires flexible translation between numerical representation (table) and graphical representation (chart), and evaluating which graphic accurately reflects the underlying data relationships.
- This is a textbook example of the cross-representational understanding emphasized in *interpret.pdf*.

🟡 4. Competency #6: Use knowledge of how the real world impacts outcomes to make contextual judgments

- Moderately relevant.
- Students must apply basic real-world knowledge about digital storage:
  - "Copying" preserves all data exactly—no loss or gain;
  - A larger empty drive will appear mostly free after copying a small dataset;
  - File types (music, photos) retain their identity and size during transfer.
- Without this knowledge, a student might wrongly assume the system "reallocates" space or compresses data.
- While less central than the top three, this real-world awareness is crucial for eliminating implausible options (e.g., charts showing very little free space).

✅ 1. 能力 #2：判断数学结果或结论在问题情境中是否合理

（Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem）

- 最高相关性。
- 学生需首先从表格中提取旧存储棒的总已用空间（如 Music + Photos = 550 + 338 = 888 MB），加上空闲 112 MB，验证总和为 1000 MB（符合 1 GB 容量）。
- 当内容被完整迁移到 2000 MB 的新盘时，数据总量不变（仍为 888 MB），因此新盘状态应为：
  - 已用：888 MB（其中 Music 和 Photos 比例不变）
  - 空闲：2000 − 888 = 1112 MB
- 四个选项中，只有正确反映“已用 ≈ 44%，空闲 ≈ 56%”且 Music/Photos 内部比例与原表一致的图表才合理。
- 学生若误认为“新盘会自动填满”或“比例会重置”，就会选错——这正是考查其能否基于情境判断图形结果是否合理。

✅ 2. 能力 #1：将数学结果还原到现实世界情境中

（Interpret a mathematical result back into the real-world context）

- 高度相关。
- 本题的核心操作是“复制全部内容到更大容量的新设备”。学生必须理解：
  - 文件不会因换盘而改变大小；
  - 新盘的“空闲空间”会显著增加；
  - 各类文件的绝对大小不变，因此它们在新盘中的相对占比会下降。
- 例如，若旧盘中 Photos 占 33.8%，在新盘中应占约 16.9%（338/2000）。学生需将这一数值关系“翻译”为视觉图表中的扇区大小。
- 这一过程正是将抽象数量关系（888 MB / 2000 MB）解释为现实设备状态（新U盘看起来“更空”）。

✅ 3. 能力 #5：以多种格式解读数学结果，并比较不同表示方式

（Interpret mathematical outcomes in a variety of formats... compare representations）

- 高度相关。
- 题目提供了表格数据（精确数值）和多个饼图选项（视觉比例）。
- 学生必须：
  1. 从表格中读取 Music、Photos、Free 的具体 MB 数；
  2. 计算它们在新盘（2000 MB）中的百分比；
  3. 将这些百分比与四个饼图的扇区大小进行视觉匹配与比较。
- 这要求在数值表示（表格）之间灵活转换，并判断哪种图形准确再现了数据关系。
- 正是 *interpret.pdf* 中强调的“跨表示形式理解与验证”的典型任务。

🟡 4. 能力 #6：利用现实世界知识对结果进行情境化判断

（Use knowledge of how the real world impacts outcomes...）

- 中等相关。
- 学生需调用关于数字存储设备的基本常识：
  - “复制”意味着数据完全保留，不增不减；
  - 更大容量的空U盘在复制小量数据后，大部分空间仍为空；
  - 文件类型（音乐、照片）不会因迁移而合并或改变属性。
- 若缺乏此常识，可能误以为系统会“重新分配”空间或压缩数据。
- 虽然不如前三项突出，但现实知识仍是排除错误选项（如空闲空间过小的图）的关键。